

Thoracic Human Vertebrae

Quantitative Three-Dimensional Anatomy

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This study details the quantitative three-dimensional surface anatomy of thoracic vertebrae based on a study of 144 vertebrae. The thoracic spine was found to have three distinct regions: upper, middle, and lower segments. The two end segments appear to be transitional zones toward cervical and lumbar regions. The middle zone (T3 to T9) is of utmost importance due to the presence of the combination of narrow spinal canal and critical vascular supply. Means and standard errors of the means for linear, angular, and area dimensions of vertebral bodies, spinal canal, pedicle, pars interarticularis, spinous and transverse processes, and rib articulations are provided for all thoracic vertebrae. This information is necessary for constructing accurate mathematical models of the human spine. It will also provide a better understanding of the spine, and allow for a more precise clinical and surgical management of spinal problems. [Key words: thoracic spine, vertebrae, anatomy, vertebral dimensions, spinal canal, pedicles]

THERE ARE EXCELLENT TEXTS describing the shapes and sizes of vertebrae in the various regions of the spine.²² However, often these are artist's drawings based on visual inspection of cadavers.^{10,11} No quantitative geometric information is provided. Such information is necessary to build an anatomic data base, which may be used, among other things, for the development of accurate and valid mathematical models of the human spine. Mathematical models depicting the thoracic spine have been used to study etiology, development, and treatment of scoliosis,²² and the mechanical response of the spine to external forces.^{2,4,14,17,24} The geometric data concerning vertebrae used in these spinal models are generally of much lower quality than the mathematical sophistication of the models themselves. As the models' quality of prediction depends on the quality of the incorporated data, it is essential that high-quality data sets should be available.

There are other important uses for a quantitative vertebral geometric data base. Physical properties of vertebrae are related to their size and shape. The size of the spinal canal has been found to be related to the incidence of low-back pain,²⁰ and size is affected by physiologic stresses in childhood.⁶ The shape of a lumbar vertebra—specifically, the concavity in the horizontal plane of its posterior wall—has been postulated to affect the frequency of disc herniation. The size and orientation of pedicles are important dimensions for the design of transpedicular fixation devices, as well as for associated surgical procedures. Several studies have been conducted in the past few years

with the primary goal of documenting the pedicle dimensions for internal fixation device application.^{5,16,23,25} These studies, however, documented relatively few thoracic vertebral levels, and not all vertebral dimensions.

The purpose of this paper is to report quantitative three-dimensional surface geometry of the thoracic spine, including clinically important dimensions (eg, linear, angular, and area measurements) from T1 to T12 vertebrae. This is the first paper in a series of three, the others covering the cervical and lumbar regions.

MATERIALS AND METHODS

Fresh autopsy whole-spine specimens, consisting of C2 to L5, were obtained. The specimens were identified by age, sex, height, weight, and cause of death (Table 1). Each vertebra was tagged to indicate spine specimen number and the vertebral level. All muscles and their attachments were removed. The ligaments, discs, and remaining soft tissues were softened by boiling. They were then subsequently immersed in bleach (0.5 hour) and 75% ethanol (1 hour), which cleaned off all the soft tissue and divided the spine into individual vertebrae. Two pointed, threaded rods, 3.0 mm in diameter and 3.2 cm in length, were threaded into the vertebral body of each vertebra. The rods were oriented so that one lay approximately in the frontal plane and the other in the mid-sagittal plane (Figure 1). Anteroposterior and lateral radiographs were taken to identify the tips of the pins in the local anatomic coordinate system of the vertebra with origin at point C, the center of the upper end-plate (Figure 1).

Each anatomic part was quantified in terms of a set of points, either spanning a surface or forming a contour (Figure 2). For example, each rib articulation was represented by up to 20 points (five rows of three to four points), and the pedicle was defined by points located circumferentially. The number of points chosen in a set was such that the surface geometry was defined reasonably well by the straight lines joining the adjacent points. Each vertebra was marked with the points to be studied. The three-dimensional coordinates of all the points were obtained with the use of the morphometer, an instrument described below. The vertebral parts whose geometric characteristics were studied are listed in Table 2, together with the nomenclature used to identify them.

The Three-Dimensional Morphometer. This specially designed instrument consisted of one linear variable differential transformer (LVDT) and two angular rotary variable differential transformer (RVDT) displacement transducers.¹⁸ The transducers were so arranged that their axes, meeting at a point, established a spherical coordinate system. The movement of the morphometer pointer tip produced voltage changes proportional to its location in space. The instrument was connected to a computer (Digital Equipment Company, Boston, Massachusetts; 11/03) through a 12-bit A/D converter to record the coordinate data.

To measure the three-dimensional location of a point on a vertebra, the following procedure was adopted, with the specimen held in the morphometer stand. First, the three tips of the threaded rods were touched in a given order, recording their spatial coordinates. Then the point to be digitized was touched and its coordinates recorded. Computer programs were developed to convert these voltages into Cartesian

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Table 1. Spine Specimen Data

Specimen No.	Age	Sex	Weight (kg)	Height (cm)	Cause of death
S3	48	M	70	178	Coronary artery disease
S4	42	F	60	167	Breast carcinoma
S7	53	M	54	167	Esophageal carcinoma
S9	19	F	68	166	Acute lymphocytic leukemia
S10	44	F	66	161	Right lung carcinoma
S13	56	M	70	157	Renal failure
S14	56	M	74	177	Pancreatic carcinoma
S15	59	M	84	171	Myocardial infarction
S20	49	F	57	157	Third degree burns
S22	53	M	68	172	Carcinomatosis
S23	57	M	75	170	Myocardial infarction
S24	20	M	68	170	Subdural hemorrhage

coordinates with respect to an axis system defined by the tips of the three pins. The largest error of the instrument in locating a point on the vertebra was ± 0.5 mm.

Data Collection and Analysis. The vertebra to be studied was rigidly secured in the morphometer stand. The three tips of the threaded pins and each of the points on the vertebra were touched with the morphometer pointer in a predetermined sequence, and their positions recorded. Some of the anatomic points (eg, the lower end-plate) could not be recorded from the initial position of the vertebra. In such a situation, the vertebra was turned upside-down and fixed in a new position. In addition to the anatomic points, the tips of the threaded pins were redigitized to allow for proper spatial orientation. The coordinate transformation, determined from the two positions of the tips of the three threaded pins, made it possible to transform the coordinates of the anatomic points in the second position to the coordinates of the anatomic points obtained in the first position.

The first step in data analysis was to transform the digitized points for each vertebra to their own local anatomic coordinate system. Points A, B, D, and E define respectively the right superior, the left superior, the right inferior, and the left inferior edges of the posterior wall of the vertebral body (Figure 1). These four points were chosen as the reference points because they were consistent and showed minimum variation in the presence of osteophytes. Using the least-squares method, the best-fit plane passing through these points was determined. This defined the frontal plane. Point C was the centroid of the superior end-plate, and formed the origin of the local anatomic coordinate system. A line perpendicular to the frontal plane formed the z-axis. A line parallel to line AB and pointing to the left formed the x-axis. Finally, the y-axis was defined as a line perpendicular to the z-x plane. The positive directions of the three axes are shown in Figure 1. Three types of anatomic parameters (linear, angular, and surface area measurements) were computed.

Linear Parameters. The necessary computer software was developed to process the coordinate data to yield well defined linear anatomic dimensions. For example, the end-plate was characterized by its width (EPW) and depth (EPD). The results of all the specimens were pooled to calculate the mean and standard error of the mean for each parameter.

Angular Parameters. Orientation of a plane was determined either by a unit vector perpendicular to the plane or by angles made with respect to each of the three anatomic planes. A plane may be defined by a minimum of three noncolinear points. Since in the present study each plane was characterized by more than three points, eg, the rib articulation was characterized by up to 20 points, the least-squares technique was used to yield the plane that represented the best fit.

To provide an alternative measure of the end-plate angles measured herein, a parallel radiographic study was conducted. The end-plate angles with respect to the transverse plane (EPI_{tu}, EPI_{tl}) were measured on ten lateral radiographs of the thoracic spine.

Area Parameters. The projected areas of the digitized part onto the three planes of the local anatomic system were computed. The surface or cross-sectional area was computed as the square root of the sum of the squares of the three projected areas. Mean and standard error of the mean were then computed.

Statistical Analysis. Statistical analysis using single-factor analysis of variance (ANOVA) and Scheffe's *F* test were used to compare the significant differences between parameter values as functions of vertebral level. A statistical significance level of 5% was used throughout.

Accuracy of Measurements. The accuracies in computing linear, angular, and area parameters were studied using an object with known values. A Plexiglas piece with sides cut at 30°, 45°, and 60° was constructed. A sheet of millimeter graph paper, with different shapes (triangles, rectangles, circles, etc.) drawn on it, was pasted on each of the cut surfaces. The pins (markers) and three reference points similar to those on a vertebra were identified on the Plexiglas piece. Thus, a pseudo-vertebra with surfaces of known inclination and parts of known dimensions and surface areas was created. The parts were digitized, using the protocol adopted for the vertebral parts. The linear, angular, and area parameters were computed and compared with the known values. The errors were found to be less than 5% in all cases.

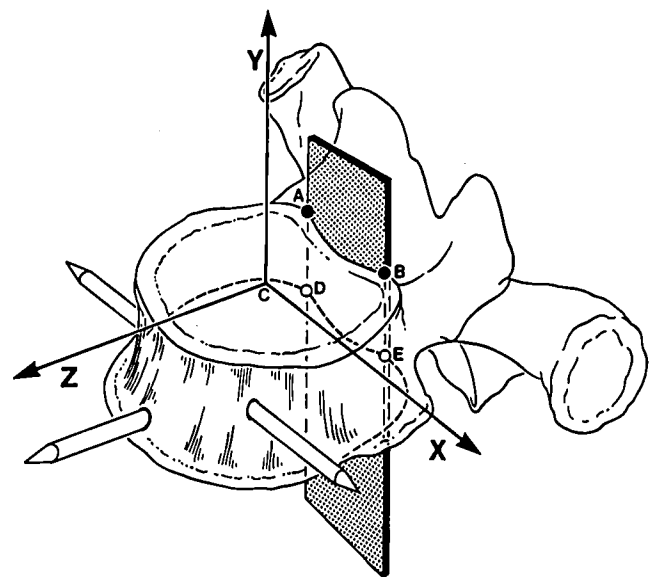


Fig 1. Pin orientation to establish a local vertebral coordinate system, with the points digitized to define the local vertebral frontal plane (A,B,D,E) and the local coordinate system (center at point C).

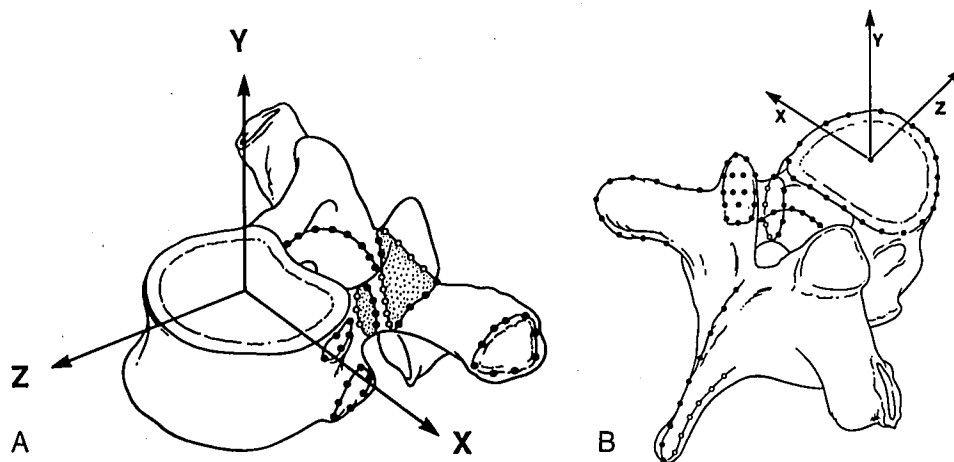


Fig 2. Points used to define various anatomic structures. **A**, Rib articulations, pars interarticularis, and pedicle. **B**, End-plate, transverse process, spinous process, pedicle, articular facet, spinal canal.

RESULTS

A total of 12 C2 to L5 spines were studied. This paper describes results from the 144 thoracic vertebrae that were examined and measured using the above protocol.

As described, the quantification of vertebral parts was done in terms of three types of parameters: linear, angular, and surface area dimensions. The complete results of the parameters measured for the thoracic vertebrae (T1 to T12) are listed in Tables 3–7. The nomenclature for the measured parameters is presented graphically in Figure 3, and also given in Table 2. The results are shown graphically as well as described below, while statistical differences are shown in the tabulated data.

Vertebral Body

The linear dimensions of the upper and lower end-plates consisted of lateral width (EPW) and anteroposterior depth (EPD). There existed a gradual increase from T1 to T12 in all four dimensions, ranging from 55% for the lower end-plate width to 73% for the upper end-plate depth (Figure 4A). The increase in width was greatest for the two distal-most vertebrae (T11 and T12) for both the upper and lower end-plates; however, for the anteroposterior diameter, the largest increase occurred proximal at T2 for the upper end-plate, and at T1 for the lower end-plate.

By observing the lateral widths of adjacent vertebrae, it was found that this dimension for the lower end-plate of the proximal vertebra was consistently larger than the corresponding dimension of the upper end-plate of the distal vertebra. This was not true for the anteroposterior diameters, which were about equal for the adjacent end-plates.

The ratio of the width to the depth gave evidence of three distinct regions within the thoracic spine (Figure 4B). For the upper end-plate, the middle region existed from T4 to T10, and for the lower end-plate, it occurred from T5 to T8. At T1, the width was 1.3 times that of the anteroposterior diameter, and at T12, the ratio was 1.2. In the middle region, the ratio was approximately one.

In relation to the total area, the upper and lower end-plates at T12 were, respectively, 3.2 and 2.7 times larger than those at T1. The increase was steady and essentially linear (Figure 4C). Using the end-plate linear dimensions of width and depth, end-plate area was estimated as both an ellipse ($A = \pi \cdot EPW \cdot EPD/4$) and a triangle ($A = EPW \cdot EPD/2$) to determine which geometric shape would best approximate the area. These values also are shown in Figure 4C.

The inclination of the upper and lower end-plates remained constant in the entire thoracic region. The average transverse plane angle of the

upper end-plate was 1.7° , while that of the lower end-plate was 2.1° . The transverse plane (wedge) angle was defined as the sum of these two angles, and was found to average 3.8° .

In the supplementary radiographic study performed, the average angular difference between upper and lower end-plates was found to be 3.9° .

The posterior height of each vertebral body was measured. There was a gradual increase in the posterior vertebral height from a value of 14.3 mm at T1 to 22.7 mm at T12, an increase of 60% (Figure 4D). This corresponds to an average increase of 0.8 mm per vertebral level.

Table 2. Table of Nomenclature (See Figure 3)

Body Part	Mnemonics	Dimension
Vertebral body	EPW	End-plate width
	EPD	End-plate depth
	EPA	End-plate area
	EPI	End-plate inclination
Spinal canal	VBH	Vertebral body height
	SCW	Spinal canal width
	SCD	Spinal canal depth
	SCA	Spinal canal area
Pedicle	PDW	Pedicle width
	PDH	Pedicle height
	PDA	Pedicle area
	PDI	Pedicle inclination
Rib articulations	PDM	Pedicle moment of inertia (2nd area)
	CTA	Costovertebral process articulation area
	CVA	Costovertebral articulation area
Pars interarticularis	PIA	Pars intra-articularis area
Spinous process	SPL	Spinous process length
Transverse process	TPW	Transverse process width
Suffixes	a	Anterior
	p	Posterior
	f	Frontal plane
	s	Sagittal plane
	t	Transverse plane
	u	Upper
	l	Lower

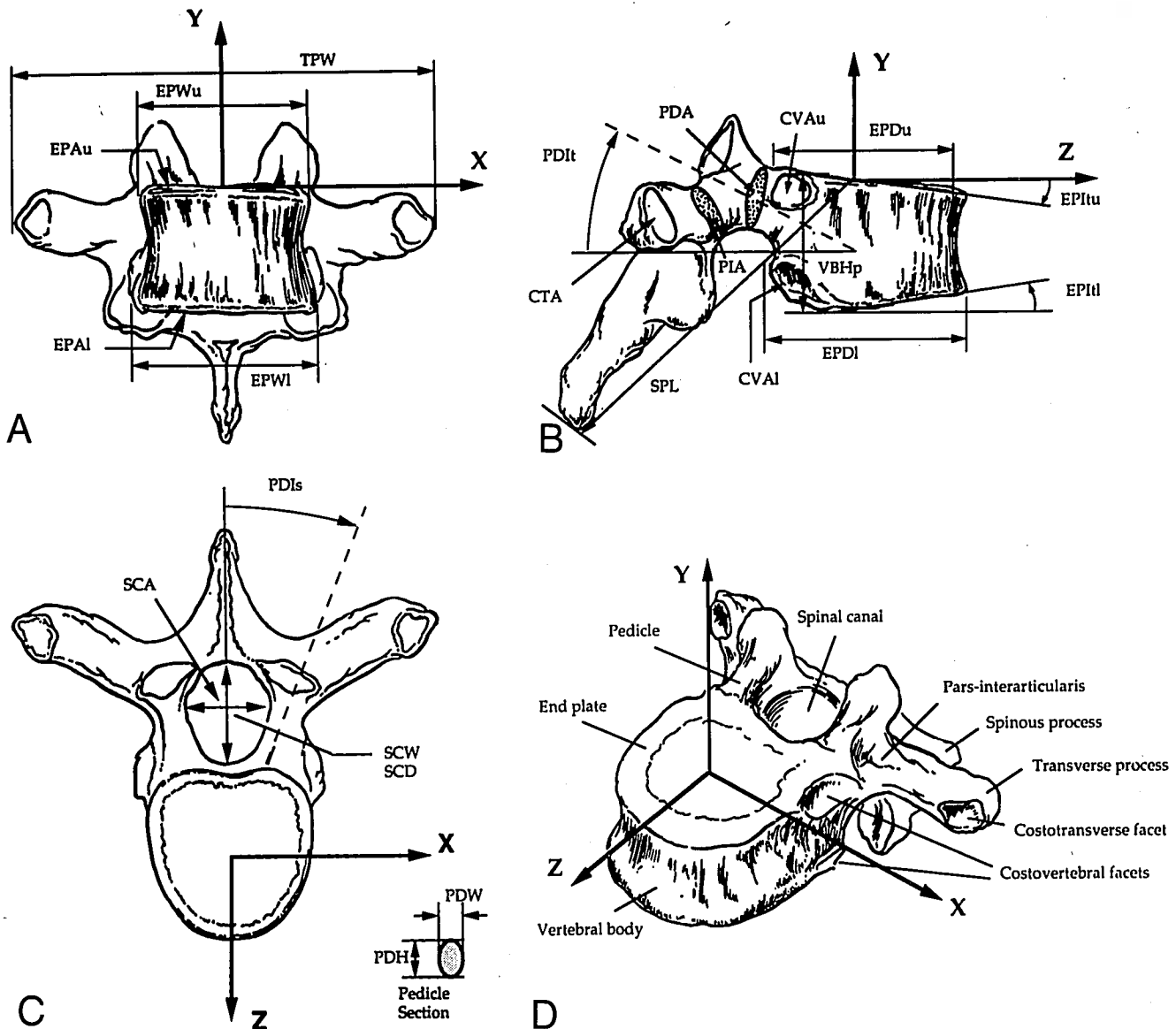


Fig 3. A–C, Three orthogonal views (front, side, and top) of a thoracic vertebra are shown. The various dimensions determined in the present study are indicated, and are further explained in the nomenclature Table 1. D, The fourth view is an isometric view showing the coordinate system used.

Spinal Canal

The linear dimensions of the spinal canal were lateral width and anteroposterior depth. There was evidence of three distinct regions in the thoracic spine with respect to the lateral dimension of the spinal canal (Figure 5A, Table 4). This curve resembled a parabola, with decreasing lateral width in the first region, leveling off in the second region, and then increasing in the third region. The narrowest portion in the lateral dimension (spinal canal width) occurred in the middle region (T4 to T9), with an average value of 17.4 mm. With respect to the anteroposterior diameter, there was less variability, with an average value of 16.2 mm. At T12, however, there was a significant increase to 18.1 mm.

The ratio between the width and depth is shown in Figure 5B. This ratio exhibited several distinct regions from T1 to T12, with the spinal canal having a ratio of 1.0 from T4 to T6.

The cross-sectional area of the spinal canal remained relatively constant between T2 and T10, averaging 198 mm² (Figure 5C). It was

larger at T1, and from T11 to T12 there was a large increase to 280 mm². The narrowest portion of the thoracic spinal canal occurred at T3, with a cross-sectional area of 189 mm². As with the end-plates, an attempt was made to model the spinal canal area both as an ellipse and a triangle. The results are shown in Figure 5C.

Pedicle

The linear dimensions of the pedicle were its cross-sectional height and width. The height averaged 9.6 mm at T1, then increased rapidly to 11.9 mm at T3 (Figure 6A). From T3 to T8, a levelling-off occurred, and the height averaged 11.9 mm. From T8 through T12, there was another linear increase of approximately 1 mm per vertebral level. The width was initially 8.4 mm at T1, then decreased to an average of 6.3 mm from T3 to T8. From T8 to T12, there was a linear increase of 0.7 mm per vertebral level. At T12, the height was 16.6 mm and the width was 8.7 mm.

Table 4. Spinal Canal Dimensions (See Figure 3 and Table 2 for Nomenclature)

	Thoracic Vertebral Level											
	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12
Liner dimensions (mm)												
SCW	21.8	19.5	18.3	17.0	17.1	17.3	17.3	17.7	17.9	18.2	19.4	22.2 (T4)
	0.71	0.34	0.54	0.57	0.68	0.74	0.83	0.63	1.19	0.78	0.95	1.12
SCD	16.4	15.3	15.9	16.2	16.3	16.5	16.1	15.9	15.7	15.5	16.0	18.1
	0.42	1.25	1.09	0.89	0.72	0.62	0.69	0.76	0.82	0.70	0.46	0.62
SCW/SCD	1.3	1.3	1.2	1.0	1.0	1.0	1.1	1.1	1.1	1.2	1.2	1.2
Cross-sectional area (mm²)												
SCA	213.0	200.0	180.0	192.0	201.0	206.0	199.0	194.0	200.0	202.0	220.0	280.0
	30.6	27.6	25.7	25.9	15.7	14.3	17.0	15.9	16.2	14.7	15.7	16.3

Note: For each parameter, the upper row shows the means, while the lower row shows the standard error of the means. For each parameter, an analysis of variance (ANOVA) was performed to determine ($P < 0.05$) which vertebral levels were significantly different (shown in parentheses).

The height-to-width ratios for the left, right, and average pedicles are shown in Figure 6B. At T1, the average ratio was 1.1, and, with progression down the thoracic spine, the ratio increased to approximately 1.9 at T5, and remained relatively constant thereafter.

The cross-sectional areas of the both left and right pedicles from T1 to T12 are shown in Figure 6C; they resemble a parabola. Initially, there was a decrease from T1 to T3, a constant value in the middle region, and a significant increase in the final region. From T8 to T11, the slope of the curve increased dramatically. The initial area was about 53 mm², decreasing to 32 mm² in the middle region, and finally reaching a value of 94 mm² at T12. The progression from T7 to T12 was approximately threefold. Again, the pedicle shape was estimated as an ellipse and a triangle, as shown in Figure 6C.

The pedicle angles were defined for its centerline (perpendicular to the pedicle cross-section) with respect to each of the two planes: sagittal and transverse. The inclination measurements of the pedicles showed a general left/right symmetry, as seen in Figure 6D. Note that there is rather a wide variation in the sagittal plane angle for the mid-thoracic vertebrae. The progression of inclination relative to the sagittal plane decreased generally from approximately 27° at T1 and T2, to 10° at T12. The inclination with the transverse plane remained relatively constant from T3 to T11, at approximately 9°. The angle jumped from 7.6° at T1 to 17.9° at T2, and from 8.7° at T11 to 5.0° at T12.

Using the measured height and width of each pedicle, the second area moments of inertia with respect to their axes were computed ($PDMs = I_{HH}^3W/64$, $PDMt = I_{HH}^3W/64$), and are shown in Figure 6E. These values are proportional to the bending stiffness of the pedicle. As shown in the Figure, each moment of inertia remained relatively constant from T1 to T8, and rose sharply to T11 with a slight drop at T12. The sagittal plane stiffness was always larger than the transverse plane stiffness.

Rib Articulations

To examine the articulation areas between the ribs and vertebrae, the areas were divided into costovertebral and costotransverse. The costovertebral area was further divided into the upper and lower costovertebral area. Several notable patterns developed: T11 and T12 were uniarticular with respect to the vertebral body. From T1 through T10, the ribs articulate with the corresponding vertebrae and one proximal biarticular (ie, rib 7 articulates with T7 and T6. Hence, from T1 through T10, each vertebral body has two articular surfaces.

The areas of the upper and lower articular surfaces were determined, and are presented in Table 6. The total costovertebral area per rib, costotransverse area, and their sum are shown in Figure 7. From T1 to T2, there was a sharp drop in the costovertebral and total areas, and then

the areas increased to T6 for both the costovertebral and total rib articulations. Both areas were relatively constant from T6 to T10, followed by a sharp decrease at T11 and T12. This sharp decline at T11 and T12 was due to the sharp drop in the costotransverse area at these levels, and to the uniarticular vertebral surfaces.

Pars Interarticularis

The pars interarticularis area (PIA) showed a gradual increase from T4 to T11 (Figure 8, Table 7). The cross-sectional area at T3 was about 90 mm², and at T11 it was 160 mm². At T12, it decreased suddenly to 98 mm² due to the inferior facets changing from the thoracic to the lumbar type.

Spinous and Transverse Processes

The spinous process length (SPL) was measured from the centroid of the superior end-plate (origin of the coordinate system) to the most inferoposterior tip. At T1, it was about 50 mm (Figure 9, Table 8). The length of the process continued to be about 50 mm until T11, where it decreased to 46 mm.

The transverse process width (TPW) was defined as the distance from the tip of the left transverse process to that of the right (Figure 9, Table 8). It displayed the three distinct regions of the thoracic spine. It decreased from T1 to T4, remained constant to T10, and decreased from T10 to T12.

DISCUSSION

There are excellent, classic anatomic texts that provide drawings depicting the thoracic vertebrae from different viewpoints; however, these are visual impressions with little quantification of the vertebral dimensions. To our knowledge, the present study is unique in that it provides, for the first time, a comprehensive quantitative three-dimensional surface anatomy of the thoracic vertebrae. Such quantified three-dimensional data are essential for the construction of accurate spinal mathematical models. Problems of spinal instability, spinal trauma, and, more recently, the design of pedicle screw fixation devices, are some clinical examples that it is hoped will benefit from this quantified anatomic data base, through the use of more precise dimensions of the spinal canal, and the development of more anatomically accurate mathematical models.

Our method used relatively simple and inexpensive equipment with high accuracy, in contrast to much more expensive techniques, such as three-dimensional computed tomography (CT) scan, which may not yield such accurate quantitative data.

Table 5. Pedicle Dimensions (See Figure 3 and Table 2 for Nomenclature)

	Thoracic Vertebral Level											
	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12
Linear dimensions (mm)												
PDWr	8.2	8.4	7.0	5.5	6.2	6.0	6.5	6.7	7.6	8.3	8.8	8.8
PDWI	0.44	1.06	0.78	0.71	0.53	0.78	0.80	0.35	0.54	0.70	0.43	0.81
PDHr	8.7	7.9	6.5	7.0	5.7	6.0	5.2	6.7	7.7	9.7 (T1)	10.7 (T5-7)	8.6
PDHI	0.53	0.97	0.55	0.49	0.38	0.87	0.45	0.63	0.65	0.78	0.84	0.68
PDH/PDWr	9.3	11.1	11.8	11.9	11.2	12.0	11.8	12.5	13.9 (T1)	14.7 (T1-2, T5)	16.9 (T1-8)	16.5 (T1-8)
PDH/PDHI	0.53	0.44	0.23	0.46	0.51	0.44	0.30	0.52	0.79	0.49	0.49	0.71
PDH/PDWr	9.9	11.6	12.0	12.2	11.4	11.6	12.2	12.5	13.8 (T1)	15.0 (T1-2, T5-6)	17.8 (T1-9)	16.8 (T1-8)
PDH/PDHI	0.41	0.30	0.32	0.50	0.36	0.47	0.27	0.41	0.55	0.24	0.29	0.94
PDH/PDWr	1.1	1.3	1.7	2.2	1.8	2.0	1.8	1.9	1.8	1.8	1.9	1.9
PDH/PDHI	1.1	1.5	1.4	1.4	2.0	1.9	2.3	1.9	1.8	1.6	1.7	1.9
Cross-sectional area (mm²)												
PDAr	52.2	46.3	38.1	32.5	31.6	33.5	36.8	43.8	52.3	64.8	88.4 (T2-8)	90.9 (T1-9)
PDAl	6.14	3.98	3.0	4.08	1.72	2.27	3.75	4.67	5.17	4.68	7.5	13.00
PDAlr	54.4	45.5	41.9	38.4	33.1	34.6	34.5	42.0	51.6	61.6	98.3 (T1-10)	97.0 (T1-10)
PDAlr	4.60	4.95	2.29	3.19	3.13	3.78	3.40	3.78	4.90	5.26	5.93	12.30
Angular dimensions (°)												
PDIsr	28.1	28.9	22.5	21.8	20.2	19.4	23.4	22.5	19.3	14.4	12.9	8.0
PDIsr	1.72	6.17	3.21	4.37	3.56	3.96	4.32	4.70	5.00	1.94	1.32	1.10
PDIsr	26.0	28.3	16.2	17.2	11.0	13.4	18.0	16.7	10.2	10.4	13.2	11.5
PDIsr	2.52	8.58	2.21	1.97	2.01	2.58	3.60	3.27	1.99	2.29	2.02	2.70
PDIsr	4.6	16.5	8.1	6.4	8.6	7.0	10.9	12.1	8.3	6.8	8.9	4.8
PDIsr	1.3	7.93	1.23	1.48	1.81	1.44	1.92	1.82	1.64	0.84	0.89	1.00
PDIsr	10.5	19.2	9.4	10.5	7.6	9.6	12.1	11.1	8.5	7.3	8.4	5.3
PDIsr	1.5	9.77	1.28	1.41	1.99	1.52	1.92	1.70	1.71	1.88	1.12	1.12

Note: For each parameter, the upper row shows the means, while the lower row shows the standard error of the means. For each parameter, an analysis of variance (ANOVA) was performed to determine ($P < 0.05$) which vertebral levels were significantly different (shown in parentheses).

Table 6. Rib and Processes Dimensions (See Figure 3 and Table 2 for Nomenclature)

Thoracic Vertebral Level												
	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12
Costovertebral articulation area												
CVAur	45.3	29.0	26.7	31.3	42.6	53.4	47.8	39.5	45.1	56.3	71.9 (T2-4)	61.0
	5.54	2.42	2.20	3.71	5.51	6.08	4.04	6.83	4.35	5.89	5.38	8.77
CVAul	56.6	29.5	34.2	36.9	41.3	51.4	53.4	44.1	40.7	51.9	60.0	47.9
	7.44	2.14	3.29	5.72	3.38	5.65	7.85	7.14	3.65	6.23	7.48	4.20
CVAir	27.9	21.0	22.5	22.7	25.2	39.4	40.7	44.5	38.7	30.0	—	—
	2.22	2.67	2.85	3.41	4.15	6.66	4.61	6.33	6.45	6.39	—	—
CVAil	19.5	12.1	19.9	20.0	25.7	33.4	30.7	25.5	26.2	26.9	—	—
	1.52	1.58	4.12	2.09	4.97	4.70	3.80	4.09	1.40	2.52	—	—
CVA	74.7	45.8	51.7	55.5	67.4	88.8	86.3	76.8	75.4	82.6	66.0	54.5
total per rib												
CTAr	49.7	49.4	61.5	61.2	51.3	48.2	50.5	63.8	60.9	62.3	16.7 (T3-4, T8-10)	—
	4.92	5.24	4.40	3.90	2.79	4.35	4.42	26.7	4.97	17.3	1.61	—
CTAI	65.0	50.8	55.9	59.2	51.3	48.7	42.3	40.0	48.6	49.3	11.7 (T1-6, T9-10)	—
	4.20	2.40	8.01	2.78	3.39	6.07	3.50	6.00	6.18	8.67	0.35	—
CVA + CTA	132.0	95.9	110.4	115.7	118.7	137.3	132.7	128.7	130.1	138.4	80.2	54.5
total per rib												

Note: For each parameter, the upper row shows the means, while the lower row shows the standard error of the means. For each parameter, an analysis of variance (ANOVA) was performed to determine ($P < 0.05$) which vertebral levels were significantly different (shown in parentheses).

Table 7. Pars Intra-Articularis, Spinous, and Transverse Processes (See Figure 3 and Table 2 for Nomenclature)

Thoracic Vertebral Level												
	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12
Cross-sectional area (mm²)												
PIAr	—	—	118.0	75.0	89.0	58.0 (T9, T11)	113.0	113.0	140.0	131.0	145.0	86.0
	—	—	1.3	2.8	10.8	9.6	4.1	4.5	19.0	13.2	32.4	3.4
PIAI	—	—	73.0	66.0 (T11)	81.0	98.0	98.0	121.0	124.0	148.0	179.0	98.0
	—	—	17.6	6.3	5.4	19.7	7.1	15.3	7.3	12.9	45.6	9.9
Linear dimensions (mm)												
SPL	50.1	52.1	51.7	51.1	52.1	53.8	50.5	52.8	51.3	49.3	45.6	47.4
	1.45	1.46	1.53	2.00	1.90	1.31	1.79	1.22	1.09	1.02	1.15	1.70
TPW	75.3 (T3-12)	69.4 (T11-12)	60.8	56.9	61.1	61.3	60.4	59.9	59.3	58.4	52.2	46.9 (T1-3, T5)
	1.74	0.90	1.54	3.77	1.90	1.75	2.34	1.92	1.46	1.48	1.28	1.84

Note: For each parameter, the upper row shows the means, while the lower row shows the standard error of the means. For each parameter, an analysis of variance (ANOVA) was performed to determine ($P < 0.05$) which vertebral levels were significantly different (shown in parentheses).

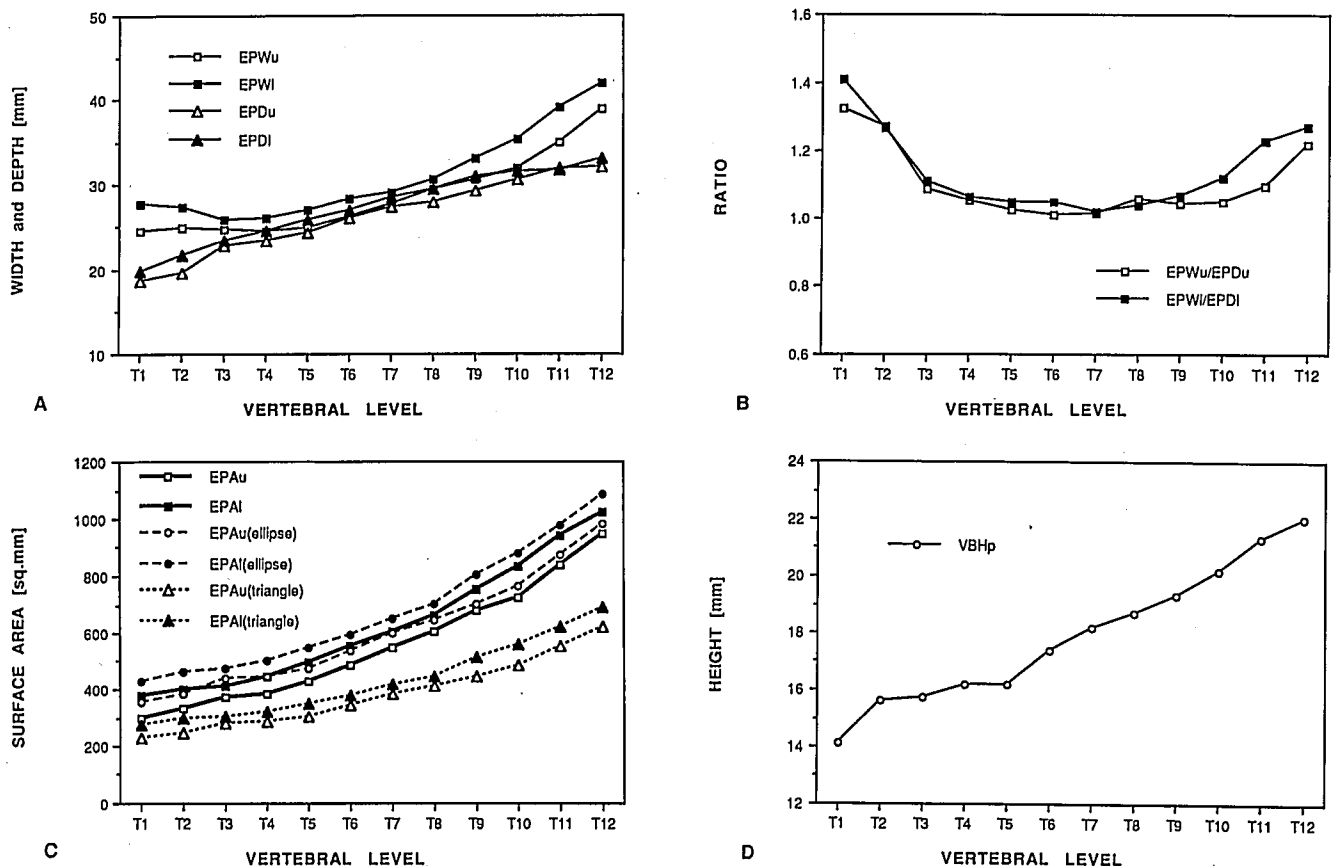


Fig 4. Vertebral end-plate dimensions as functions of vertebral levels T1 to T12 are shown. **A**, Linear dimensions of the upper and lower end-plates. **B**, Ratio of end-plate width to depth for the upper and lower end-plates. **C**, Surface areas of upper and lower end-plates are indicated by whole lines. Dashed lines depict the end-plate areas, assuming the end-plate shape to be either an ellipse or a triangle. **D**, Heights of the posterior wall of the vertebral body.

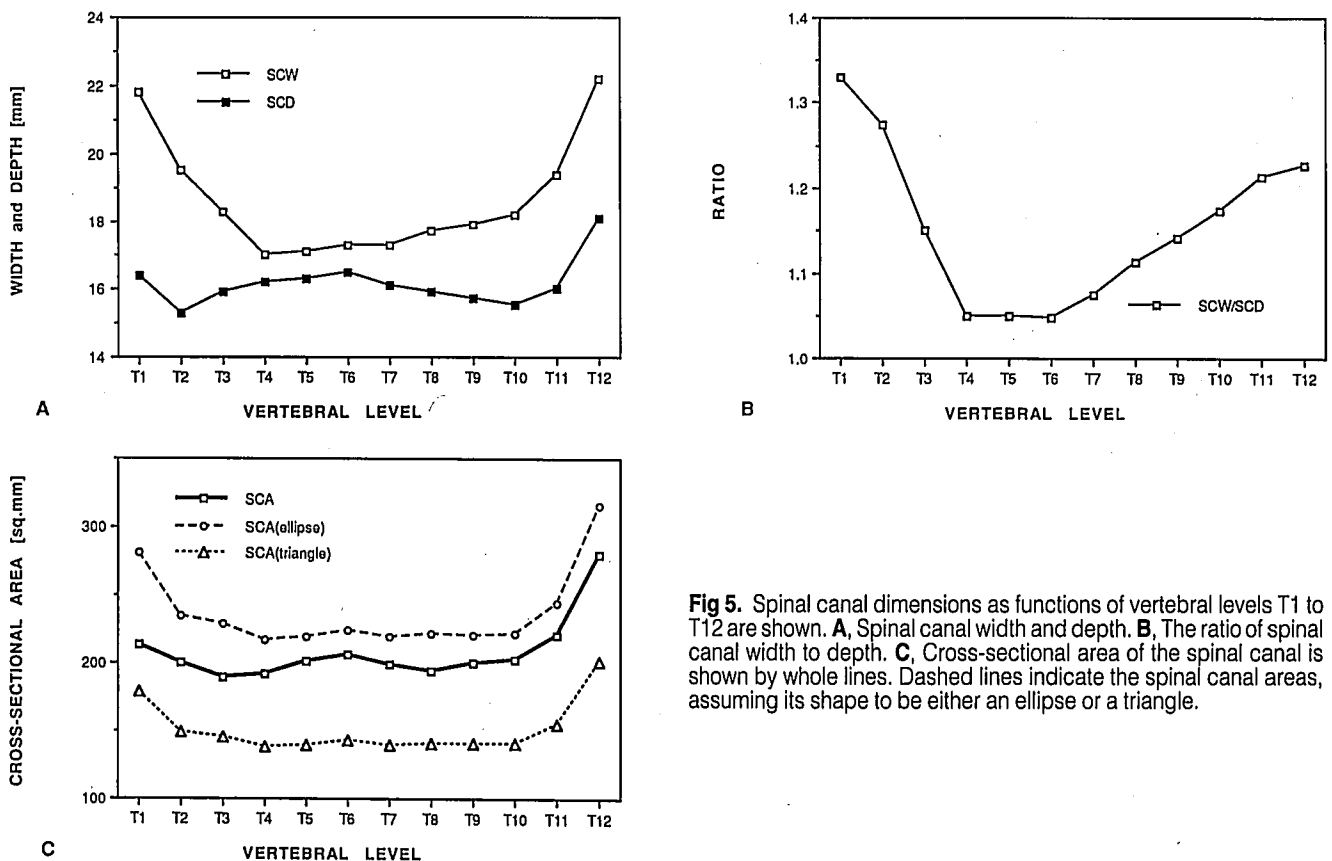


Fig 5. Spinal canal dimensions as functions of vertebral levels T1 to T12 are shown. **A**, Spinal canal width and depth. **B**, The ratio of spinal canal width to depth. **C**, Cross-sectional area of the spinal canal is shown by whole lines. Dashed lines indicate the spinal canal areas, assuming its shape to be either an ellipse or a triangle.

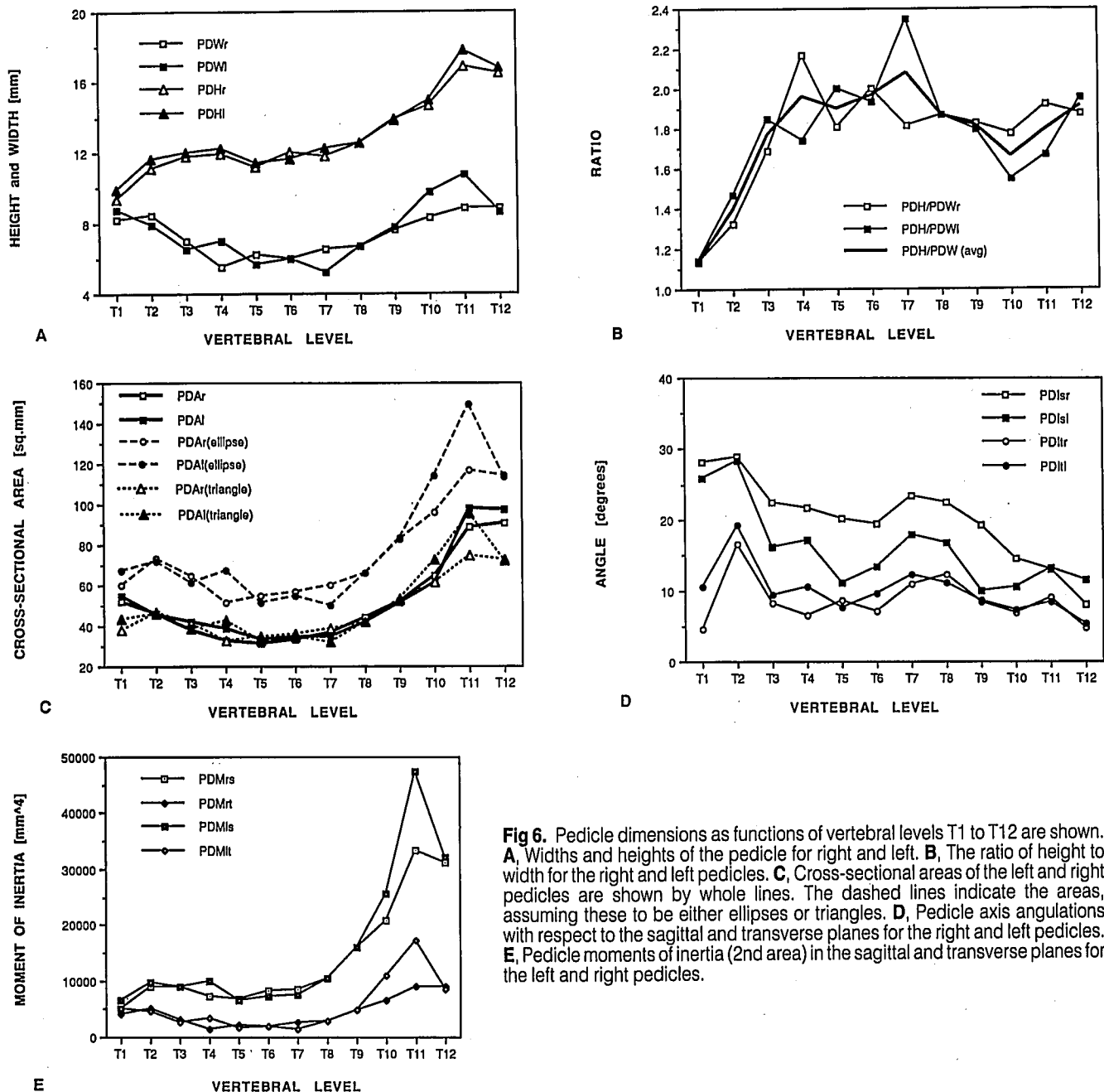


Fig 6. Pedicle dimensions as functions of vertebral levels T1 to T12 are shown. **A**, Widths and heights of the pedicle for right and left. **B**, The ratio of height to width for the right and left pedicles. **C**, Cross-sectional areas of the left and right pedicles are shown by whole lines. The dashed lines indicate the areas, assuming these to be either ellipses or triangles. **D**, Pedicle axis angulations with respect to the sagittal and transverse planes for the right and left pedicles. **E**, Pedicle moments of inertia (2nd area) in the sagittal and transverse planes for the left and right pedicles.

Transition Zones

From the data presented herein, the thoracic spine may be naturally divided into three distinct regions. These are the transition from cervical to thoracic; the transition from thoracic to lumbar; and a middle region. The idea of three thoracic regions is not new. Among others, Grant's Anatomy (1972), Gray's Anatomy (1973), Hollingshead (1969), and Parke (1982) have noted these three regions. However, these studies are not in agreement regarding which vertebrae are contained in each region.

The upper region, from cervical to thoracic, was characterized best by a narrowing of the end-plate and spinal canal width from T1 to T4, while the depths remained relatively constant. As a result, the width-to-depth ratios decreased from T1 to T4, thus identifying this upper transition zone.

The middle thoracic region was characterized by a relatively narrow end-plate and spinal canal from T4 through T9 or T10. A small spinal canal makes this region susceptible to cord impingement, although rib articulations add significant stiffness to the region.³ The middle region (T4 through T9) has been called the *critical vascular zone* for the spinal cord.^{9,21} Thus, the middle region has both the narrowest spinal canal and it corresponds almost exactly with the part of the spinal cord to which the blood supply is least profuse. Numerous clinical articles have expressed the warning that strict care must be taken when surgery is contemplated in this region.

The final transition zone, from thoracic to lumbar, was most evident through a distinct enlarging of the spinal canal area from T10 to T12. This occurs when the ribs go from a complete unit at T10 to floating ribs at T11 and T12 (ie, no ventral bony attachment), thus providing less

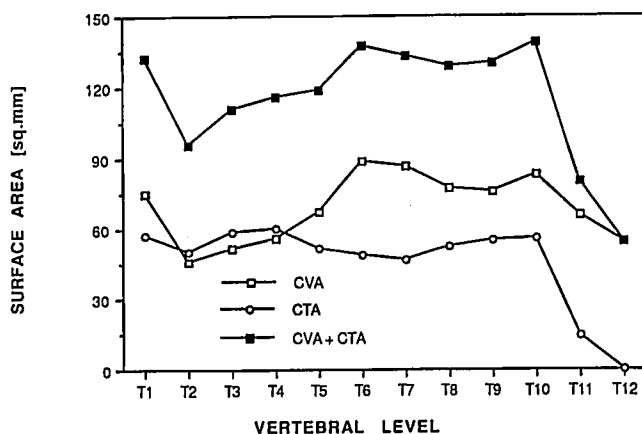


Fig 7. Rib articulation areas as functions of vertebral levels T1 to T12. The areas articulating with the vertebral body, transverse process, and the total area are shown.

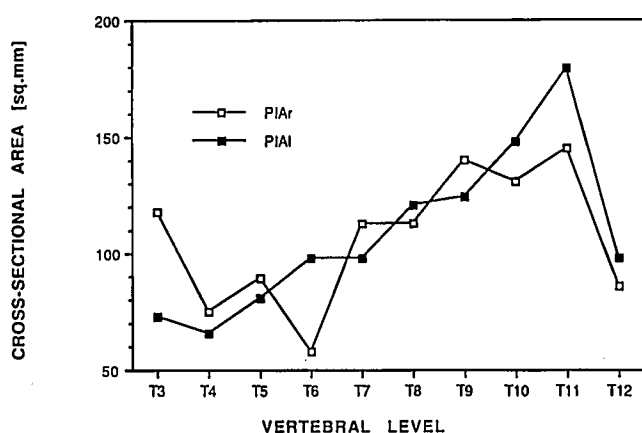


Fig 8. Pars interarticularis areas as functions of vertebral levels T1 to T12 on the left and right are shown.

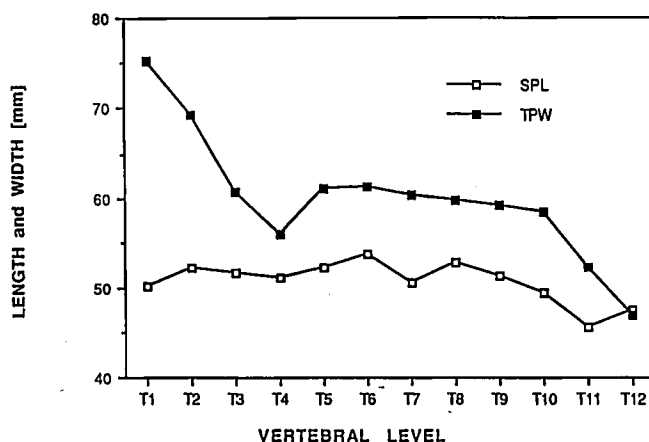


Fig 9. Spinous and transverse process dimensions as functions of the vertebral levels T1 to T12.

stiffness to the vertebral column in this region. This transition was observed in the decreased rib articulation areas with regard to both the vertebral body and the transverse processes. There was also corresponding increase in the end-plate width, especially the upper end-plate.

Kyphotic Curve

The question of the etiology of the dorsal curve (anterior concavity) was addressed as early as 1879, when Aeby¹ stated that the bony configuration of the vertebral body accounted for this curve. Thirty years ago, Hollingshead described the overall embryologic spinal curves as initially being totally C-shaped, with anterior concavity.¹² He explained that at birth the cervical spine developed an anterior convexity secondary to the response of the paraspinal cervical musculature; later, the development of the psoas muscle gave anterior convexity to the lumbar spine. Hence, it would appear that the dorsal anterior concavity in the thoracic spine is present due to phylogeny, while the cervical and lumbar curves are adaptive.^{8,12} In our study, we found a vertebral wedge angle of 3.8° per vertebra, giving a total of approximately 45.6° for the entire thoracic spine; this is the bony angle. These data are supported by our additional radiographic study, in which we found the average bony angle per thoracic vertebra to be 3.9°. We did not examine or quantitatively measure the intervertebral disc angle or height, so we cannot comment on the total etiology of the curve.

Previous Studies

Several quantitative anatomy studies have been completed recently for the thoracic and lumbar vertebrae.^{5,7,16,23,25} In two of the studies,^{16,25} the focus lay solely on the pedicle dimensions, which are clinically important for design and surgical implantation of internal fixation devices. In the remaining studies, anatomic measurements were made on only two or three selected thoracic vertebrae. Table 8 provides a comparison between the various parameters measured in the present study and those in some of the earlier work.

The first parameter that we compared was the anteroposterior depth of the lower end-plate. Our results corresponded well to those of Berry et al,⁵ but were consistently 4–5 mm greater than those of Scoles et al²³ and Cotterill et al.⁷ Although not stated in either article, the reported anteroposterior depth may be at the body mid-height, which would lead to consistently smaller values. Our lower end-plate widths agreed best with the results of Scoles et al, but were generally several millimeters less than the data of Berry et al, and slightly larger than those of Cotterill et al. The posterior vertebral body heights reported herein were very close to those given by Cotterill et al, and were consistently 1 to 2 mm less than those reported by Berry et al and Scoles et al.

The spinal canal depths in the present study agree well with the data of Berry et al⁵ and Scoles et al.²³ The spinal canal widths reported by Scoles et al are consistently less than those in our study by approximately 1 mm.

The pedicle heights of this study agree well with the results of Berry et al,⁵ Zindrick et al,²⁵ and Scoles et al.²³ The average pedicle widths in this study were consistently 1 to 2 mm greater than in the earlier work. However, there was some disparity between the three studies. The pedicle widths given by Scoles et al were 2 to 3 mm less than those of Zindrick et al for T3, T6, and T9, and 3 to 4 mm less than our results and those of Krag et al,¹⁶ in studies of the same levels. There is significant variation between all studies in the angulation of the pedicles with respect to both the sagittal and transverse planes. In the studies by Berry et al and Zindrick et al, a negative pedicle angle with respect to the sagittal plane was found, but this was not observed in the present study or in the study by Scoles et al. The pedicle angles with respect to the

Table 8. Comparison of Our Results with Those of Previous Studies (Average Values of Selected Linear and Angular Dimensions Compared)

Parameter	Level	Present Study	Berry (1987)	Zindrick (1987)	Scoles (1988)	Krag (1986, 1988)	Cotterill (1986)
Linear dimensions (mm)							
EPDI	T1	19.7	—	—	15.5	—	—
	T2	21.6	19.0	—	—	—	—
	T3	23.2	—	—	18.7	—	—
	T4	24.5	—	—	—	—	—
	T5	25.8	—	—	—	—	—
	T6	26.9	—	—	23.7	—	21.8
	T7	28.5	28.0	—	—	—	—
	T8	29.4	—	—	—	—	—
	T9	31.0	—	—	27.4	32.5	—
	T10	31.6	—	—	—	32.0	—
	T11	31.8	—	—	—	31.0	—
	T12	33.4	31.2	—	28.8	32.0	27.5
EPWI	T1	27.8	—	—	26.4	—	—
	T2	27.4	33.5	—	—	—	—
	T3	25.9	—	—	27.6	—	—
	T4	26.0	—	—	—	—	—
	T5	27.0	—	—	—	—	—
	T6	28.2	—	—	28.7	—	25.1
	T7	29.1	33.2	—	—	—	—
	T8	30.5	—	—	—	—	—
	T9	33.0	32.5	—	—	—	—
	T10	35.4	—	—	—	—	—
	T11	39.1	—	—	—	—	—
	T12	42.1	46.8	—	41.7	—	37.3
VBHp	T1	14.1	—	—	16.8	—	—
	T2	15.6	16.5	—	—	—	—
	T3	15.7	—	—	18.6	—	—
	T4	16.2	—	—	—	—	—
	T5	16.2	—	—	—	—	—
	T6	17.4	—	—	19.8	—	17.5
	T7	18.2	19.1	—	—	—	—
	T8	18.7	—	—	—	—	—
	T9	19.3	—	—	21.8	—	—
	T10	20.2	—	—	—	—	—
	T11	21.3	—	—	—	—	—
	T12	22.7	24.8	—	25.9	—	23.6
SCD	T1	16.4	—	—	15.2	—	—
	T2	15.3	15.0	—	—	—	—
	T3	15.9	—	—	15.6	—	—
	T4	16.2	—	—	—	—	—
	T5	16.3	—	—	—	—	—
	T6	16.5	—	—	15.7	—	14.5
	T7	16.1	16.6	—	—	—	—
	T8	15.9	—	—	—	—	—
	T9	15.7	—	—	15.4	—	—
	T10	15.5	—	—	—	—	—
	T11	16.0	—	—	—	—	—
	T12	18.1	17.2	—	17.9	—	14.6
SCW	T1	21.8	—	—	21.2	—	—
	T2	19.5	18.3	—	—	—	—
	T3	18.3	—	—	17.4	—	—
	T4	17.0	—	—	—	—	—
	T5	17.1	—	—	—	—	—
	T6	17.3	—	—	16.5	—	15.1
	T7	17.3	17.1	—	—	—	—
	T8	17.7	—	—	—	—	—
	T9	17.9	—	—	16.8	—	—
	T10	18.2	—	—	—	—	—
	T11	19.4	—	—	—	—	—
	T12	22.2	20.2	—	19.9	—	19.5
PDH (avg. left and right)	T1	9.6	—	9.9	9.2	—	—
	T2	11.4	11.8	12.0	—	—	—
	T3	11.9	—	12.4	11.8	—	—
	T4	12.1	—	12.1	—	—	—
	T5	11.3	—	11.9	—	—	—
	T6	11.8	—	12.2	11.5	—	—
	T7	12.0	12.0	12.1	—	—	—
	T8	12.5	—	12.8	—	—	—
	T9	13.9	—	13.8	12.9	—	—
	T10	14.9	—	15.2	—	—	—

Table 8. Comparison of Our Results with Those of Previous Studies (Average Values of Selected Linear and Angular Dimensions Compared) (Continued)

Parameter	Level	Present Study	Berry (1987)	Zindrick (1987)	Scoles (1988)	Krag (1986, 1988)	Cotterill (1986)
PDW (avg. left and right)	T11	17.4	—	17.4	—	—	—
	T12	16.7	17.1	15.8	16.0	—	—
	T1	8.5	—	7.9	7.3	—	—
	T2	8.2	6.2	7.0	—	—	—
	T3	6.8	—	5.6	3.9	—	—
	T4	6.3	—	4.7	—	—	—
	T5	6.0	—	4.5	—	—	—
	T6	6.0	—	5.2	3.5	—	4.1
	T7	5.9	5.0	5.3	—	—	—
	T8	6.7	—	5.9	—	—	—
	T9	7.7	—	6.1	3.9	6.9	—
	T10	9.0	—	6.3	—	7.5	—
Angular dimension (°)	T11	9.8	—	7.8	—	7.8	—
	T12	8.7	7.7	7.1	7.4	7.6	7.7
PDI _s	T1	27.1	—	26.6	29.8	—	—
	T2	28.6	23.0	19.1	—	—	—
	T3	19.4	—	14.6	15.3	—	—
	T4	19.5	—	12.6	—	—	—
	T5	15.6	—	9.4	—	—	—
	T6	16.4	—	9.6	10.2	—	—
	T7	20.7	7.5	8.7	—	—	—
	T8	19.6	—	8.1	—	—	—
	T9	14.8	—	7.6	9.2	6.0	—
	T10	12.4	—	4.6	—	6.0	—
	T11	13.1	—	1.2	—	0.0	—
	T12	9.8	-3.0	-4.2	9.5	0.0	—
PDI _t	T1	7.6	—	12.6	—	—	—
	T2	17.9	137	17.5	—	—	—
	T3	8.8	—	17.3	—	—	—
	T4	8.5	—	16.3	—	—	—
	T5	8.1	—	15.0	—	—	—
	T6	8.3	—	15.0	—	—	—
	T7	11.5	110	15.7	—	—	—
	T8	11.6	—	16.6	—	—	—
	T9	8.4	—	16.0	—	—	—
	T10	7.1	—	16.8	—	—	—
	T11	8.7	—	15.4	—	—	—
	T12	5.1	20	11.6	—	—	—

transverse plane by Berry et al appeared to be grossly incorrect (eg, at T2, Berry: 137.0°, Zindrick: 17.5°, present study: 17.9°).

Geometric Shape Approximations

As an aid to mathematical modeling of the spine, it is useful to be able to approximate various vertebral components by simple geometric shapes. In this study, the inferior and superior end-plates, spinal canal, and pedicles were approximated by ellipses and triangles based on their measured widths and depths or heights.

For both the lower and upper end-plates, the elliptical approximation was the best with an error of less than 10%, while the triangular approximation was a 20% underestimate at T1, increasing to a 40% underestimate at T12.

For the spinal canal, the triangular approximation was better at T1, while the elliptical approximation was better at T2 through T12. At all levels, the ellipse was an overestimate, while the triangle was an underestimate.

For the pedicle cross-section, the triangular approximation was excellent for levels T2 through T12. At T1, the measured areas were about midway between the triangular and elliptical approximations.

SUMMARY

A comprehensive data set has been presented, providing quantitative anatomy of thoracic vertebrae. We have attempted to make a "blueprint" of the external geometry of the thoracic vertebrae. Such information is essential for the development of advanced and valid mathematical models of the spine. The latter may be used to predict spine behavior (eg, development of scoliosis) and effects of surgery. The data may also be useful directly in clinical situations, such as instability determination and the design of internal spinal devices, both functional and fixation types.

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